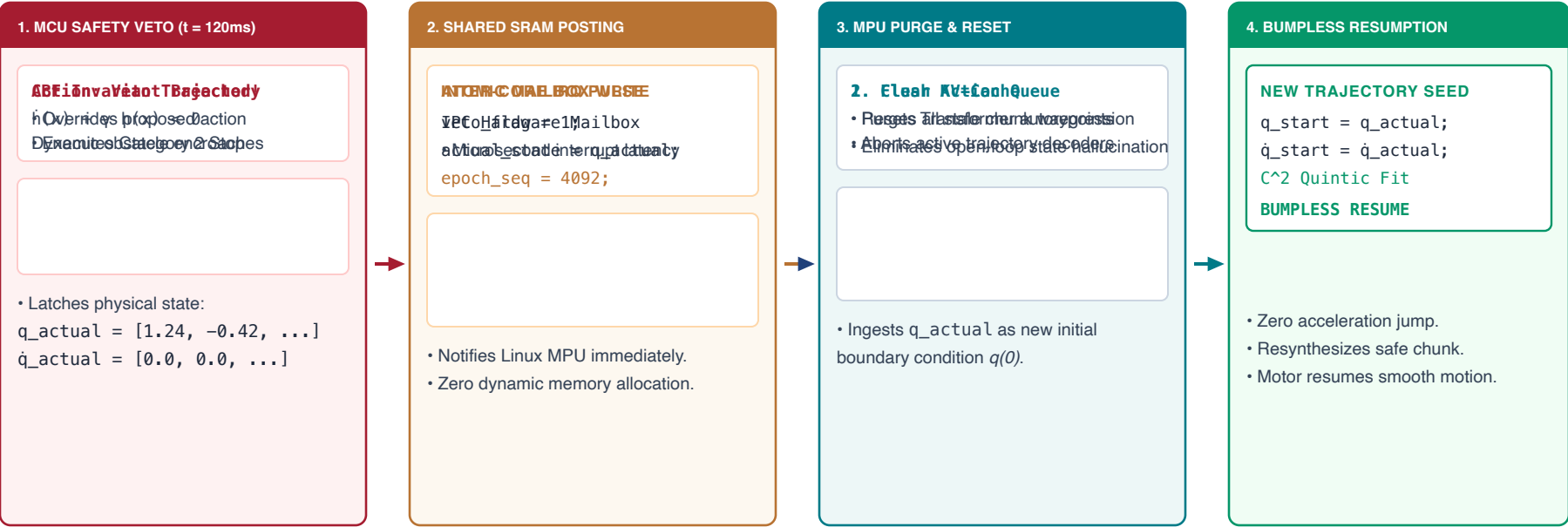


ASYNCHRONOUS STATE RE-ANCHORING HANDSHAKE UNDER SAFETY INTERVENTION

Surviving real-time MCU safety vetoes: atomic mailbox notification, queue purge, KV-cache reset, and bumpless restart



THE PHYSICAL AI STATE RE-ANCHORING CONTRACT

- 1. The Hallucination Danger:** If an MPU planner generates trajectories assuming an un-vetoed path, its next chunk commands a massive step jump from the actual physical stopping point.
- 2. Strict Hardware Handshake:** By enforcing that every new chunk starts at the exact measured physical state (q_actual , $\dot{q}_actual$), position error is zeroed.
- 3. Governed Recovery:** The system gracefully recovers from dynamic obstacles without rebooting the robot or entering an unrecoverable fault state.